

Name: _____ Date: _____

Period: _____

Keep in Mind

This is a practice test. It mirrors the real Unit 8 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

- | | |
|---------------------------------------|---|
| 1. ReLU C | (A) The building block $\text{ReLU}(A\mathbf{v} + \mathbf{b})$: multiply by weights, add a bias, then rectify. |
| 2. Neural network layer A | (B) The step size s in gradient descent — too small crawls, too big overshoots. |
| 3. Piecewise linear function F | (C) The rectifier $\max(0, x)$: keep positive inputs, send negatives to 0. |
| 4. Convolution filter D | (D) A small weight pattern slid across the input to detect one feature everywhere. |
| 5. Weight sharing G | (E) The rule $\mathbf{w} \leftarrow \mathbf{w} - s \nabla L$: step opposite the slope to shrink the loss. |
| 6. Gradient descent E | (F) A function built of straight pieces joined at folds (a sum of shifted ReLUs). |
| 7. Learning rate B | (G) Reusing one filter at every position, so a giant layer needs only a few weights. |
| 8. Covariance matrix H | (H) The symmetric matrix of variances and covariances; its eigenvectors are the principal components. |

Part B — Multiple Choice (12 pts)

Circle the best answer.

- ReLU(-3) equals (A) -3 (B) 0 ← (C) 3 (D) 1
- One neural-network layer computes (A) $A\mathbf{v}$ (B) $\text{ReLU}(A\mathbf{v} + \mathbf{b})$ ← (C) $\mathbf{v} + \mathbf{b}$ (D) $A^{-1}\mathbf{v}$
- A piecewise linear function built from N folds (kinks) has (A) $N - 1$ pieces (B) N pieces (C) $N + 1$ pieces ← (D) $2N$ pieces
- Sliding one small filter across the whole input is convolution; reusing that filter everywhere is (A) **weight sharing** ← (B) the bias (C) the loss (D) a rotation
- Gradient descent updates a weight by stepping (A) along the slope (B) **opposite the slope (downhill)** ← (C) to a random weight (D) to 0
- A positive covariance $\sigma_{xy} > 0$ between two features means they (A) move in opposite directions (B) **tend to move together** ← (C) are unrelated (D) are equal

Teacher Note

Item 1: ReLU zeroes out negatives, so $\text{ReLU}(-3) = 0$. Item 2: a layer is $\text{ReLU}(A\mathbf{v} + \mathbf{b})$ — weight, bias, rectify. Item 3: N folds cut the line into $N + 1$ straight pieces. Item 4: reusing one filter everywhere is weight sharing (why a conv layer has so few weights). Item 5: the slope points uphill, so you step the opposite way (downhill). Item 6: a positive covariance means the two features rise and fall together.

Part C — Short Answer & Computation (35 pts)

Show your work.

- One layer $\text{ReLU}(A\mathbf{v} + \mathbf{b})$. $A\mathbf{v} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$; **add the bias:** $A\mathbf{v} + \mathbf{b} = \begin{bmatrix} 4 \\ -2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$; **then ReLU keeps positives and zeroes the negative:** $\text{ReLU}\begin{bmatrix} 4 \\ -1 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}$.
- Tent function $F(x) = \text{ReLU}(x) - 2\text{ReLU}(x - 2) + \text{ReLU}(x - 4)$. **(a) Turn on only the ReLUs whose fold is left of x :** $F(0) = 0$, $F(2) = 2 - 0 + 0 = 2$, $F(4) = 4 - 2(2) + 0 = 0$, $F(6) = 6 - 2(4) + 2 = 0$. **(b) Folds at $x = 0, 2, 4$ (where each ReLU switches on); the peak is $(2, 2)$ — the graph goes up, then down, then flat (a tent).**
- Edge filter $[-1, 1]$ across $\mathbf{v} = [2, 2, 2, 6, 6]$. **(a) Each output is (next – current):** $2-2$, $2-2$, $6-2$, $6-6$, **so the feature map is** $[0, 0, 4, 0]$. **(b) The one nonzero value (4) sits at the jump from 2 to 6 — that is the edge. On the flat stretches the two window values are equal, so next – current = 0.**
- Convolution matrix $A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}$. **(a)** $A \begin{bmatrix} 1 \\ 1 \\ 6 \\ 6 \end{bmatrix} = \begin{bmatrix} 1-1 \\ 6-1 \\ 6-6 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix}$ — **the same slide-the-filter output.** **(b) The filter uses only 2 weights (the -1 and 1 , reused down every diagonal), while a full 3×4 layer would have its own weight in each entry: 12.**
- Gradient descent on $L(w) = (w - 4)^2$, $L'(w) = 2w - 8$, $s = 0.25$, $w_0 = 0$. **(a)** $w_1 = 0 - 0.25(-8) = 2$; $w_2 = 2 - 0.25(-4) = 3$; $w_3 = 3 - 0.25(-2) = 3.5$. **(b) Losses** $L(0) = 16$, $L(2) = 4$, $L(3) = 1$, $L(3.5) = 0.25$ — **shrinking fast.** **(c) It is heading toward $w = 4$, the bottom of the bowl; you know to stop when the slope $L'(w) = 0$ (no downhill left).**
- One gradient step on $L = (w_1 - 1)^2 + (w_2 - 3)^2$. $\nabla L(0, 0) = (2(0 - 1), 2(0 - 3)) = (-2, -6)$. **Step against it:** $(0, 0) - 0.25(-2, -6) = (0.5, 1.5)$. **The loss bottoms out at $(w_1, w_2) = (1, 3)$, where both slopes are 0.**
- Mean, variance, covariance, PCA for $(2, 4), (4, 2), (6, 8), (8, 6)$. **(a) Mean** $= (\frac{2+4+6+8}{4}, \frac{4+2+8+6}{4}) = (5, 5)$. **(b) Centered x :** $-3, -1, 1, 3$ **and y :** $-1, -3, 3, 1$. **Variances** $\sigma_x^2 = \frac{9+1+1+9}{4} = 5$, $\sigma_y^2 = 5$; **covariance** $\sigma_{xy} = \frac{(-3)(-1)+(-1)(-3)+(1)(3)+(3)(1)}{4} = \frac{12}{4} = 3$. **(c)** $V = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}$. **(d) Eigenvalues** $\lambda = 5 \pm 3 = 8, 2$; PC_1 (eigenvector $\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$) **holds** $\frac{8}{8+2} = \frac{8}{10} = 80\%$ **of the total variance.**

Part D — Extended Response (10 pts)

Explain your reasoning in full sentences.

- How a network learns. **(a) A layer $\text{ReLU}(A\mathbf{v} + \mathbf{b})$ multiplies the input by weights A , shifts by the bias $\mathbf{b}$ (which places each fold), then rectifies with ReLU (which zeroes negatives, creating a kink). Each row adds one shifted ReLU, so a wider A stacks more folds; a sum of shifted ReLUs is a continuous piecewise-linear function whose bends can be positioned and tilted to trace the data.** **(b) Gradient descent starts from some weights and repeatedly steps opposite the slope of the squared-error loss L (the rule $\mathbf{w} \leftarrow \mathbf{w} - s \nabla L$), which lowers L each step — it rolls downhill in the loss bowl. The bottom of the bowl is where the slope is 0: no downhill direction remains, so those weights give the smallest loss and are the ones the network learns.**

2. Why the covariance matrix unlocks PCA. (a) V is symmetric because the covariance of x with y equals the covariance of y with x ($\sigma_{xy} = \sigma_{yx}$), so the off-diagonal entries match and $V = V^T$. By the spectral theorem (Unit 6) a symmetric matrix always has *real* eigenvalues and *perpendicular* eigenvectors. (b) Those eigenvectors are the principal components — the perpendicular directions the data cloud spreads along — and each eigenvalue is the variance (amount of spread) along its direction, largest first. The total variance is $\sigma_x^2 + \sigma_y^2$, the trace of V ; since a symmetric matrix's trace equals the sum of its eigenvalues, the total spread is split exactly among the principal directions.

Teacher Note

Scoring Part D (5 pts each). D1: 3 pts (a) layer = weight/bias/ReLU, wider A = more folds = flexible piecewise-linear fit; 2 pts (b) gradient descent steps against the slope to shrink L , slope 0 = best weights. D2: 3 pts (a) $\sigma_{xy} = \sigma_{yx} \Rightarrow V$ symmetric $\Rightarrow$ real eigenvalues, perpendicular eigenvectors; 2 pts (b) eigenvectors = principal components (directions of spread), eigenvalues = variances, trace = total variance = sum of eigenvalues. Accept equivalent wording throughout.